A Hybrid LLM-Based Architecture for Multilingual Emergency SMS Triage in Sri Lanka

***Abstract*—**Six language styles co-occur in Sri Lankan emergency SMS traffic: English, Sinhala, Tamil, Singlish, Tanglish, and script mixed with English in the same message. Hand-written rule-based NLP is brittle in this setting. We present a hybrid architecture in which a large language model (LLM) is called only where rule-based and statistical methods are demonstrably insufficient; every other pipeline stage stays deterministic, auditable, and free to run. Four structured fields—incident type, location, people count, resources needed—are extracted from raw distress messages, every LLM call is wrapped in an evidence-grounded hallucination guard, and dispatch priority is assigned using the Medical Priority Dispatch System (MPDS) rather than an arbitrary point scale. Beyond prompting alone, we report five measurable novelty contributions: a Noise Taxonomy; an Adaptive Router that cuts LLM calls by 58.3% with no accuracy loss; a continuous hallucination Risk Score; a 500-example hallucination-detection benchmark comparing three detection methods; and a Cross-Model Verification study quantifying a specific reliability gap in location extraction. A fine-tuned mT5-small model is evaluated in parallel as an independent extraction method. A second fine-tuning target, a lightweight emergency-relevance classifier, rejects non-emergency traffic without hardcoded rules and without an extra inference-time LLM call. Iterative adversarial and live-conversation testing surfaced and closed a series of concrete failure modes that isolated unit testing alone would not have caught.

***Keywords*—**emergency SMS triage, low-resource multilingual NLP, large language models, hallucination detection, MPDS, hybrid architecture, code-mixed text, Sinhala, Tamil

I. Introduction

Flood, landslide, and fire emergencies recur across Sri Lanka, and plain SMS remains the most widely available channel for reporting them. Two difficulties compound when such messages are triaged automatically. First, the reported text is frequently noisy—typos, missing grammar, duplicated characters, emoji. Second, it can arrive in any of six overlapping styles: English, native-script Sinhala, native-script Tamil, Singlish (romanized Sinhala), Tanglish (romanized Tamil), or Sinhala/Tamil script mixed with English in the same message.

An earlier version of this system relied on hand-written rule-based NLP—fixed keyword lists, a small custom spelling-correction dictionary, and separately trained classifiers (TF-IDF with Logistic Regression, Random Forest, SVM, and Naive Bayes) for incident, people-count, and resource-type prediction. Our own testing across the six code-mixed styles found this approach brittle: fixed vocabulary could not keep pace with the spelling variation, elision, and code-switching that real Singlish and Tanglish messages exhibit, which motivated a redesign around LLM prompting.

Replacing every stage outright with an LLM call raises an obvious objection, though: if each agent in a multi-agent pipeline just calls an LLM, the multi-agent framing adds nothing over a single call. Our response is a *hybrid architecture*. The LLM is invoked only where it is genuinely necessary—free-text extraction from messy, code-mixed input, distress detection, and location-specificity judgments—while question generation, memory tracking, answer parsing, escalation logic, and priority scoring stay fully deterministic.

II. Related Work

Hallucination detection via natural language inference has been proposed as a general faithfulness check for ungrounded LLM output [8], typically applied to open-domain generation and summarization. Our contribution is a live, per-call deployment of the same entailment principle inside a real-time SMS pipeline, validated against a purpose-built adversarial benchmark (Section IV-D) rather than a static, general-domain corpus.

Grounded, evidence-based generation is likewise an active concern in retrieval-augmented generation research, which distinguishes an answer that is merely correct from one that is actually attributable to its source [9]. Our evidence-quote guard applies the same distinction to structured field extraction rather than open-ended text generation, and verifies each citation programmatically rather than relying on human or model judgment alone.

Script sensitivity in Sinhala language processing has recently been benchmarked directly, showing that model performance can degrade by orders of magnitude between Unicode, romanized, and mixed-script input regardless of model size [10]—a finding consistent with the language-detection and template-resolution failures we encountered and fixed during development (Section III). The broader disparity in NLP tooling and evaluation coverage across low-resource languages [11] is the structural motivation for treating Singlish and Tanglish as first-class inputs throughout our pipeline rather than as translation targets to be normalized away.

LLM-based triage classification has been studied directly for online patient inquiries using few-shot prompting [12]. Our system differs by keeping a deterministic protocol (MPDS) as the primary decision authority and using the LLM only as a bounded safety-net override—a design choice

motivated by the dangerous under-triage failure we document in Section VI, rather than by a general preference for rule-based methods over learned ones.

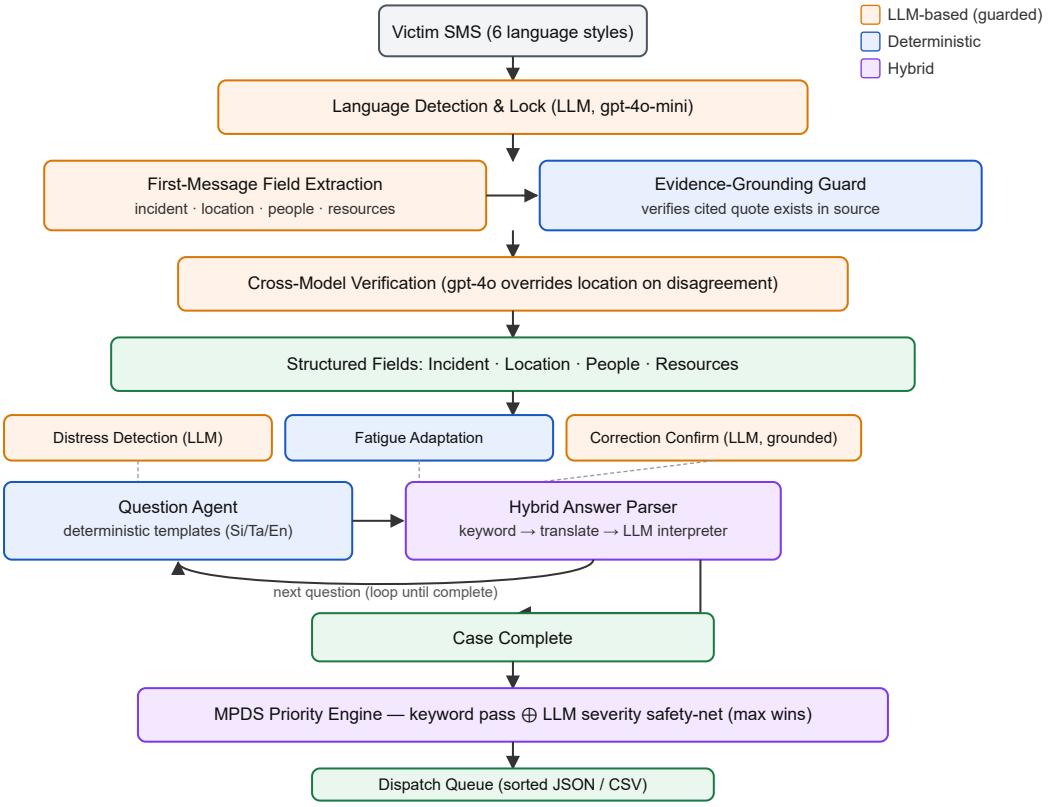


Fig. 1. System architecture. Orange = LLM-based stages, each wrapped in an evidence-grounding guard; blue = fully deterministic stages; purple = hybrid stages that fall back to an LLM only when a deterministic method fails.

III. Hallucination Guards

An LLM can invent information that was never present in the source message, which is a serious risk once the output feeds a real dispatch decision. For that reason, every pipeline stage that lets the LLM rewrite or extract text sits behind a verification guard, and nothing from that stage is trusted until it passes.

The extraction guard requires the model to cite a verbatim quote justifying every extracted field, then independently checks that the quote actually exists in the source text before the value is accepted. The grammar-fix guard works differently: it separates safe grammatical glue (articles, pronouns, copulas) from fact-bearing content words, and only rejects a correction when it introduces a new content word that is not grounded—even fuzzily—in the original text. We ended up reusing this same grounding principle across the distress detector, the location extractor, and the answer interpreter. Each of these three was built separately, and each independently hit the identical failure mode—accepting an LLM's claim without checking it against the source—before the shared grounding check was applied to all of them.

A controlled test caught two implementation bugs early on: a language-confusion issue between Tanglish, Tamil, and mixed script, and a stray-quote parsing bug in the grammar guard. Both were fixed before any production data reached the pipeline. Once fixed, the grammar-fix guard went on to catch four qualitatively distinct real LLM failure modes in the live dataset, summarized in Table I.

TABLE I. Observed LLM Failure Modes

Input	Output	Failure type	Risk
“Kandy”	“Kandy is in danger.”	Fabricated fact	60.2
Tanglish text	translated to Tamil script	Unwanted translation	65.1
fragment	meta-commentary leak	Task non-compliance	96.1
“save us” (Sinhala)	“take drums”	Semantic corruption	55.8

IV. Five Novelty Contributions

A. Noise Taxonomy

Every message is automatically tagged with the type(s) of noise it exhibits. The tags come from diffing consecutive pipeline stages, so no additional LLM call is needed to produce them. The headline finding is that broken or missing grammar affects 83.7% of the dataset, well above spelling errors (37.2%) or code-mixing (40.4%). English-language messages, somewhat surprisingly, had the *highest* grammar-noise rate of any language category at 93.1%—which pushes back on the assumption that multilingual noisy-text cleanup is mostly a typo problem.

B. Adaptive Router

Before any LLM call is spent, a free, LLM-free heuristic predicts which of three cleaning stages—spelling, emoji/duplicate removal, grammar repair—a given message actually needs. The router is a genuine hybrid in its own right: it calls established libraries (`symspellpy`, `emoji`) where they already exist, and falls back to a corpus-bootstrapped fuzzy match for Singlish/Tanglish, since no established spelling library covers code-mixed Sinhala or Tamil at all. Against Noise Taxonomy ground truth (Table II), it cuts LLM calls from 2592 down to 1080, a 58.3% reduction, without losing accuracy on the stages that matter most.

TABLE II. Adaptive Router Validation

Stage	Precision	Recall	F1
Spelling fix	0.92	0.68	0.78
Emoji/duplicate fix	0.98	0.48	0.65
Grammar fix	0.82	0.89	0.85

C. Continuous Risk Scoring

We also upgraded the binary accept/reject guards to a continuous 0–100 hallucination-risk score, combining text similarity, word-growth ratio, and a count of ungrounded new words. Doing this exposed a problem with the original binary guard: of the 135 messages it had flagged, only 4 turned out to be genuinely high-risk. The rest were harmless grammar additions that an overly strict rule had caught by mistake.

D. Hallucination Detection Benchmark

To compare detection methods properly, we built a 500-example labeled benchmark (250 hallucinated, 250 safe) and tested three approaches on it (Table III). Natural Language Inference (NLI) entailment came out strongest, once it was correctly implemented with proper premise/hypothesis framing. It was not correct on the first attempt: an earlier NLI implementation scored near zero accuracy simply because it was never given the original source text to compare against. We report that failure alongside the fix, since it is evidence of real iterative debugging rather than a single attempt that happened to work.

TABLE III. Hallucination Detector Comparison

Method	Precision	Recall	F1
Content-word overlap	0.85	1.00	0.92
Embedding similarity	0.63	0.97	0.77
NLI entailment	**0.95**	0.99	**0.97**

E. Cross-Model Verification

We extracted fields independently with `gpt-4o-mini` and `gpt-4o` and measured how often they agreed, field by field (Table IV). Location was the outlier: only 4.5% raw agreement between the two models. The cheaper model kept over-extracting generic nouns—"house", "kitchen", "street"—as if they were place names, a reliability gap that single-model evaluation alone would never have surfaced. That finding is what motivated the live cross-model verification step now in the deployed pipeline, where the larger model's verdict overrides the smaller model's whenever the two disagree.

TABLE IV. Cross-Model Field Agreement

Field	Agreement
Incident	80.6%
Location	**4.5%**
People	59.3%
Resources	76.9%

V. Fine-Tuning

We tried self-serve fine-tuning access on three platforms, in order: OpenAI (the platform is discontinued for new organizations), Google Gemini (blocked at the account level by quota/billing), and Azure OpenAI (institutional access was never confirmed). None worked out. With those external routes closed, we fine-tuned `google/mt5-small`—a small, open-weight multilingual model covering 101 languages, Sinhala and Tamil included—on Google Colab's free GPU tier using Hugging Face `transformers`. A first attempt at LoRA (parameter-efficient) fine-tuning produced a reproducible NaN-loss failure under this environment's package versions; switching to full-parameter fine-tuning fixed it and trained correctly.

On held-out validation data the fine-tuned model reached 82.76% incident-classification accuracy but only 40% location recall. That gap traces back to a documented property of the data itself: only about 5% of the 864-message training corpus has a labeled location at all, even after 8× oversampling of the location-positive examples. We treat strong performance on the well-represented field alongside weak performance on the sparse one as a coherent, honestly-reported result rather than something to gloss over, and for that reason the fine-tuned model runs in the live pipeline only as a logged parallel comparison, not as the authoritative extraction source.

A second fine-tuning target came up independently: a lightweight binary classifier deciding whether a first message is emergency-relevant at all. The point was to decline clearly unrelated traffic—a food order, a greeting—without either an extra per-message LLM call or a hand-written keyword list, and the keyword-list option was ruled out empirically, not just on principle: “help me please” (a genuine, if vague, emergency) and an unrelated message both come out of the existing extractor with a null incident field, so emptiness alone cannot tell the two apart. This classifier reuses the same `mT5-small` fine-tuning pipeline, retargeted to a single-token output (“emergency” / “not_emergency”) and trained on a purpose-built, class-balanced dataset spanning all six language styles.

VI. MPDS-Based Priority Dispatch

We considered two mass-casualty triage standards for assigning dispatch priority. START (Simple Triage and Rapid Treatment) is the more globally recognized of the two, but three of its four core checks—walking ability, breathing rate, capillary refill—need an in-person physical assessment, which SMS text simply cannot provide. The Medical Priority Dispatch System (MPDS) is built for exactly the opposite situation: remote triage from a reported description with nobody physically present, so it was the structurally correct fit here. We implement one chief-complaint protocol per extracted incident type, each of which assigns a priority code on MPDS's own scale (ECHO, DELTA, CHARLIE, BRAVO, ALPHA, OMEGA) and returns the specific textual evidence that triggered it.

An initial 26-example validation benchmark scored 100%, which turned out to be a red flag rather than good news—it was circular, since the keyword lists and the test examples drew on the same vocabulary. A second, adversarial benchmark reworded the same scenarios to strip out every literal keyword while keeping the underlying severity identical (“cannot breathe” became “I can't get any air into my lungs,” for instance). Accuracy dropped to 57.7%, with 11 dangerous under-triage cases, which is a fairly direct demonstration that keyword-only triage does not generalize past its fixed vocabulary. We added an independent LLM severity check as a safety net after that, combined with the keyword pass by always taking whichever of the two verdicts is more urgent, since under-triage is the failure direction that actually costs lives. That combination raised paraphrase-benchmark accuracy to 73.1% and cut dangerous misses from 11 down to 1 (Table V)—better, though we report the one remaining miss honestly rather than call the problem solved.

TABLE V. MPDS Adversarial Validation

Configuration	Exact match	Dangerous misses
Circular benchmark	100%	0
Paraphrase, keyword-only	57.7%	11
Paraphrase, keyword+LLM	**73.1%**	**1**

VII. Human-Factors Contributions

Two of our contributions are aimed at the victim's cognitive and emotional state rather than at language understanding as such. Fatigue detection combines three deterministic signals drawn from recent replies—reply length, presence of urgency words, and consecutive failed parses—and switches the incident, people, and resource questions from open text to numbered multiple choice once it detects sustained struggle. Distress detection runs an LLM check, grounded with the same evidence-quote mechanism described earlier, before any reply is parsed as a field answer at all; that ordering matters, because it stops a panicked or off-topic reply (a bare “please,” say) from ever being silently recorded as if it were a literal answer. A 24-example labeled benchmark spanning all six language styles reaches F1 = 1.00 now, though the first version of the detector over-flagged short-but-correct answers and had to be corrected.

Live-conversation testing later turned up a related but more general failure pattern. A fast, substring-based keyword matcher built for the people-count field was matching the English word “we” inside the unrelated Singlish word “wela,” which silently recorded a wrong headcount from what was actually a plain distress message. The same bug showed up again for the pronoun “I” inside unrelated sentences. Both were fixed by switching from substring matching to whole-word matching. A separate live finding showed hedged, non-exact counts—“I think about 3 people”—being recorded as if they were exact. We added an LLM-judged confidence check for this, deliberately not keyword-based so it generalizes by meaning across all six language styles rather than matching specific English hedge words, which asks a single confirming question before an approximate count is finalized but never blocks progress if the victim simply cannot give a more precise number.

VIII. Discussion and Limitations

Location extraction stays weak, but for a genuine data-scarcity reason rather than a modeling failure, and the MPDS severity check—substantially improved by the LLM safety net—still has one identified adversarial under-triage case we have not closed. The full set of live-pipeline novelty integrations adds real per-message latency; that is a deliberate trade-off appropriate to a research prototype, not something we would carry into a production deployment unchanged. The Sinhala and Tamil keyword translations used in the priority-triage protocols have not yet had a native-speaker review pass, unlike the question templates, which were reviewed and corrected. Concurrent multi-victim SMS traffic is also not supported yet, since conversation state currently lives in a single shared object rather than per-conversation storage; we treat this as immediate future work, ahead of any real-carrier SMS integration.

IX. Conclusion

This work shows that a defensible hybrid architecture—LLM usage confined to tasks with a demonstrated rule-based failure, hallucination guards wrapped around every generative call, and a real dispatch standard rather than an ad hoc score—can support multilingual emergency SMS triage across six Sri Lankan language styles. Five measurable novelty contributions, two independently fine-tuned models, and a documented history of adversarial testing that actually found and closed real failure modes give us reasonable grounds to call the system ready for continued evaluation toward real carrier SMS deployment.

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